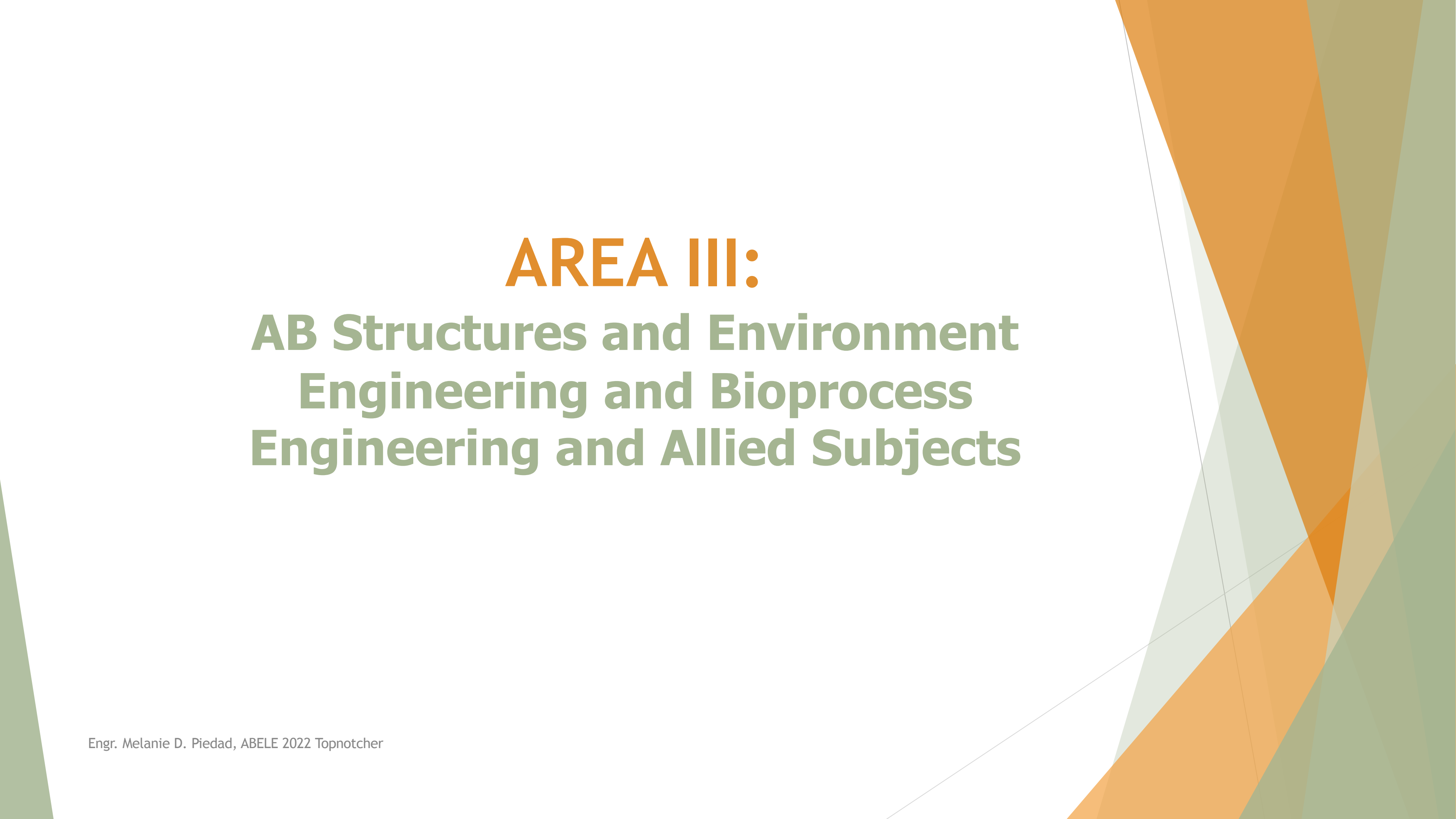




AREA III:

AB Structures and Environment Engineering and Bioprocess Engineering and Allied Subjects

Thermodynamics

Thermodynamics is the study of energy, heat, work, and their transformations.

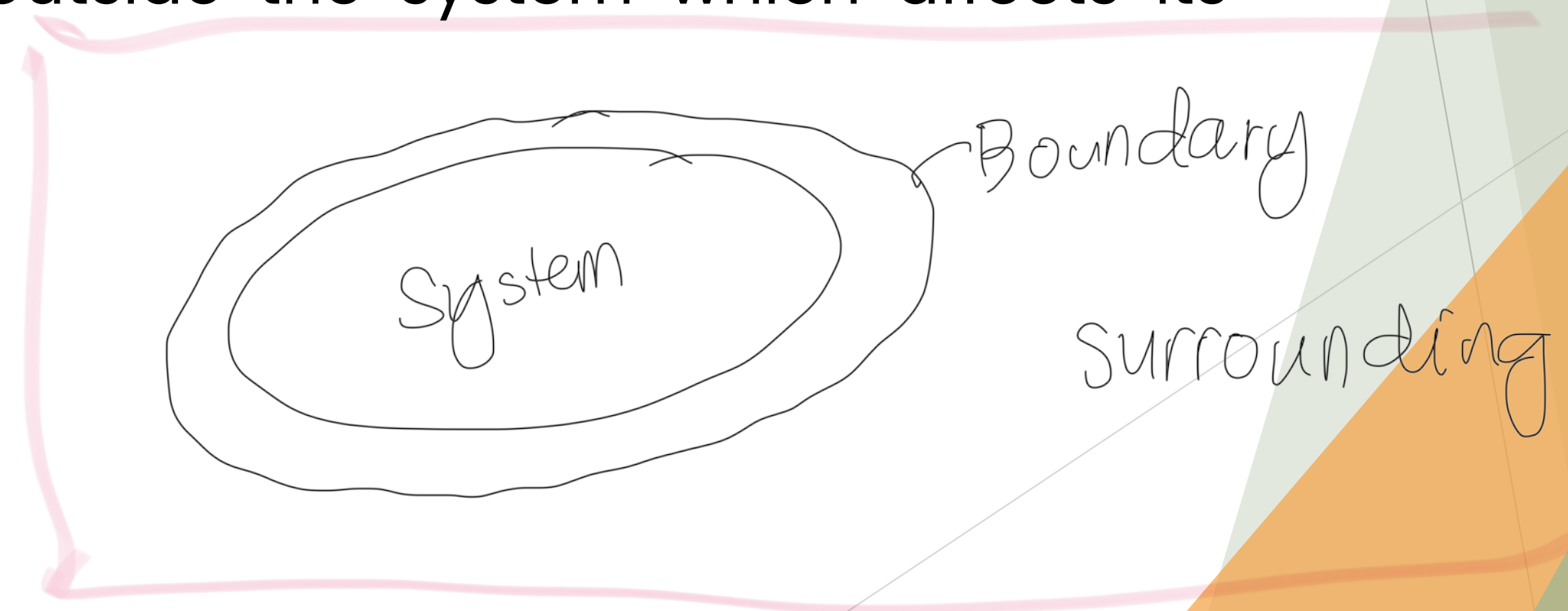
In ABE, it is crucial for designing systems like **dryers, refrigeration units, bioenergy systems, and environmental control structures.**

Thermodynamic Systems and Control Volumes

System is a definite space or area on which the study of energy transfer and energy conversion is made.

Boundary is what separates the system and the surrounding. It may be fixed, movable, or imaginary. It will not occupy any volume or mass in space.

Surrounding is anything outside the system which affects its behavior.



Thermodynamic Systems and Control Volumes

Control Volume – a specified large number thermal device has mass flow in and out of the system.

Control Surface – both mass and energy can cross the boundary of a control volume.

control vol $\rightarrow$ property: control surface

Thermodynamic Systems and Control Volumes

Open System – the mass as well as the energy may take place between system and its surrounding.

Example: internal combustion engines, air compressor, water pump, steam engine, boiler, turbine

Closed System – the mass within the boundary of the system remains constant and only the energy transfer may take place between the system and its surrounding.

Example: Pressure cooker, gas confined between a piston and cylinder

Thermodynamic Properties

Thermodynamic Property – any measurable or observable characteristics of the substance when the system remains in equilibrium state (i.e., pressure, temperature, density, volume, energy, specific volume).

Intensive Property – one whose value does not depend on the mass of the system (i.e., temperature, pressure, density, specific volume, etc.).

Extensive Property – one whose values depend on mass of the system (i.e., volume, total energy)

Thermodynamic State, Process, and Cycle

Types of equilibrium:

1. **Thermal equilibrium:** a system that features spatially-uniform temperature.
2. **Chemical equilibrium:** a system free from chemical reactions.
3. **Mechanical equilibrium:** a system has no tendency for a change of pressure over time.
4. **Phase equilibrium:** a system consisting of a mixture of multiple phases, such as liquid water and water vapor, if the composition of the mixture remains constant over time.

Thermodynamic State, Process, and Cycle

The state of a system in equilibrium is called **equilibrium state**.

A system may change from one state to another state through a **process**.

If all states in a process are equilibrium states, the process is called **quasi-equilibrium process**.

If a system undergoes a series of processes and finally returns to its initial state, we say that the system completes a **cycle**.

Thermodynamic State, Process, and Cycle

Related terms:

Adiabatic: no heat exchange (e.g., sudden expansion in dryers)

Isothermal: constant temperature

Isobaric: constant pressure (e.g., atmospheric drying)

Isochoric: constant volume (e.g., rigid containers in storage) / isometric

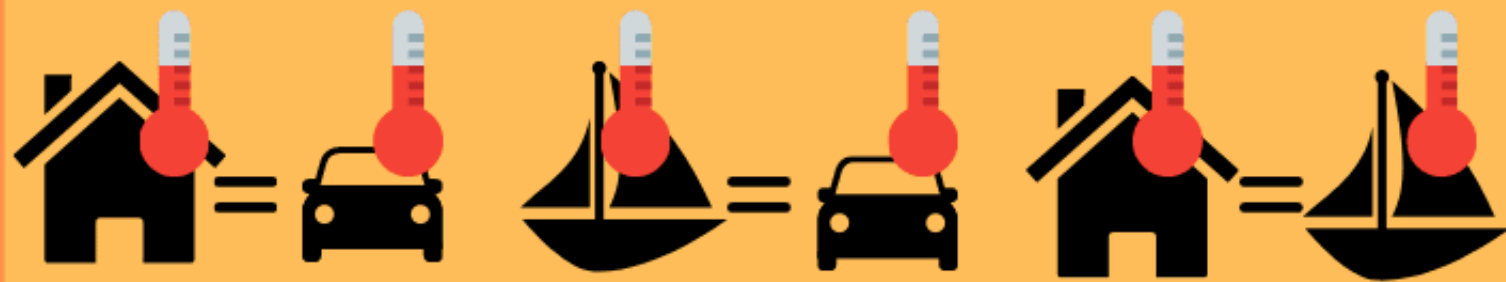
Isentropic: the entropy remains constant in the process.

Zeroth Law of Thermodynamics

The Zeroth Law states that if two systems are each in thermal equilibrium with a third system, they are also in thermal equilibrium with each other.

This allows for the use of thermometers to compare temperatures and is foundational in climate control in greenhouses or cold storage.

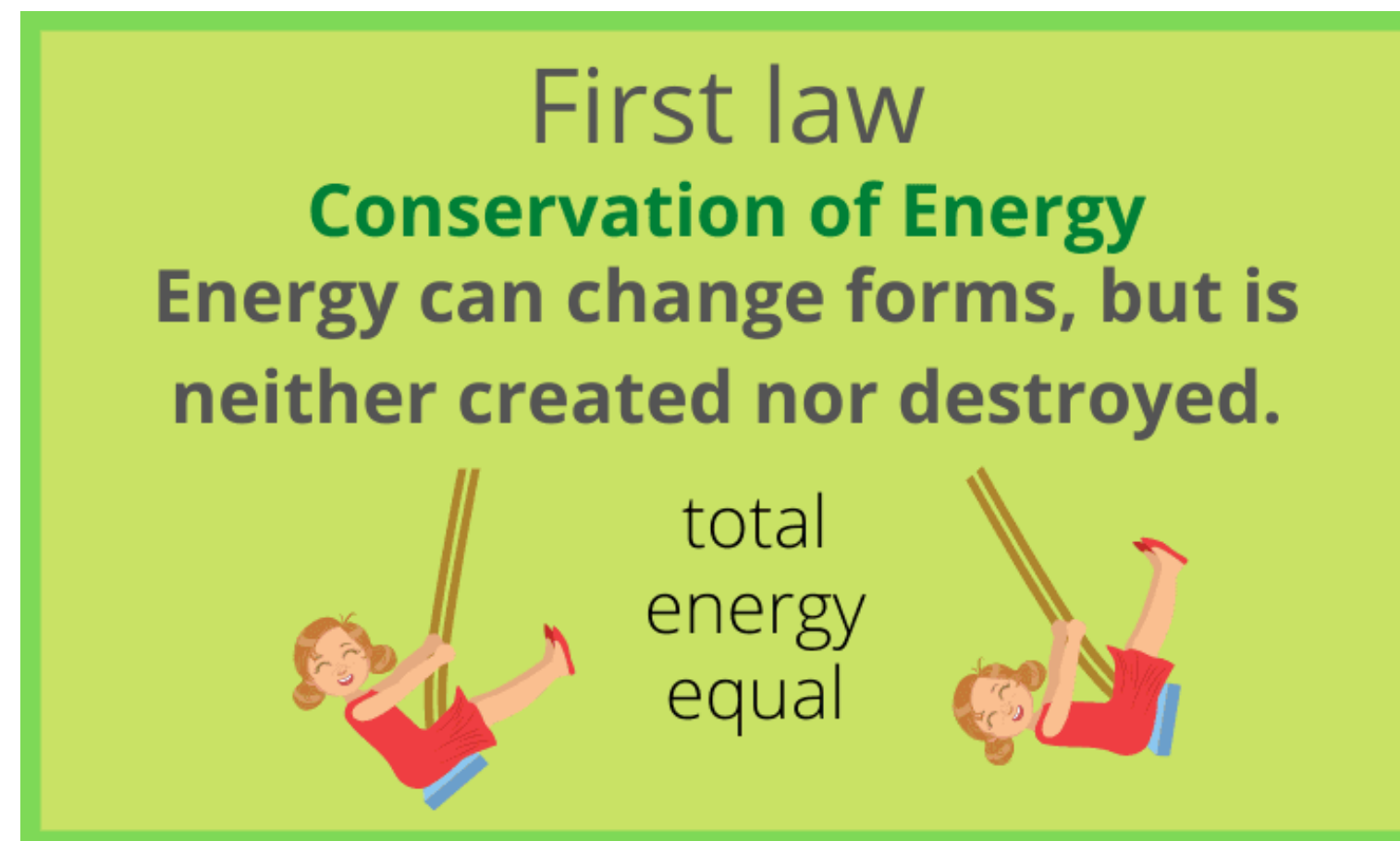
Zeroeth law
Temperature
Two systems in equilibrium with a
third system are in thermal
equilibrium with each other.



First Law of Thermodynamics

The First Law of Thermodynamics, also known as the law of energy conservation, explains that the total energy of an isolated system is constant.

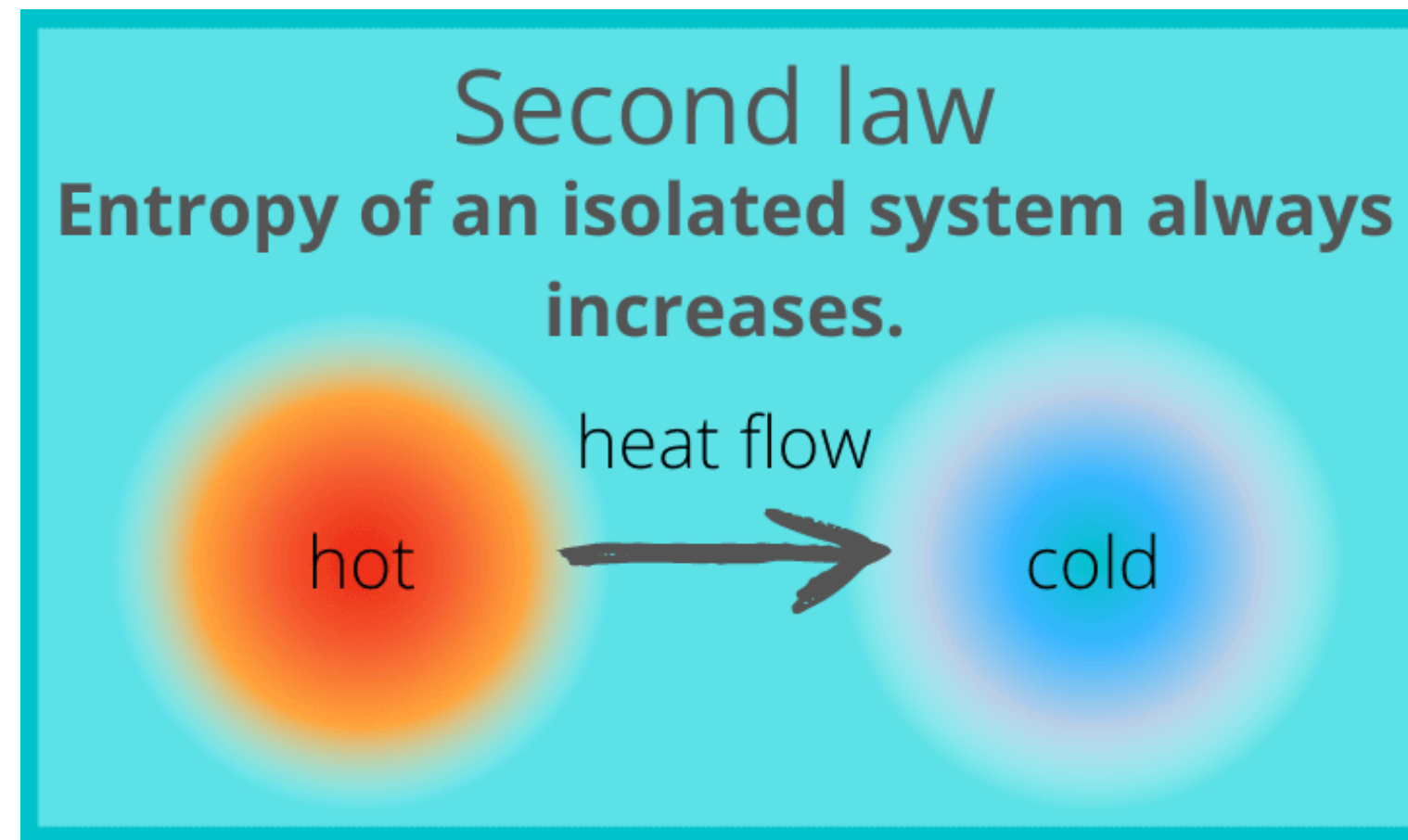
In agricultural systems, this principle is applied in machines and dryers where mechanical or electrical energy is transformed into heat energy.



Second Law of Thermodynamics

The Second Law of Thermodynamics introduces the concept of entropy, which measures the disorder in a system.

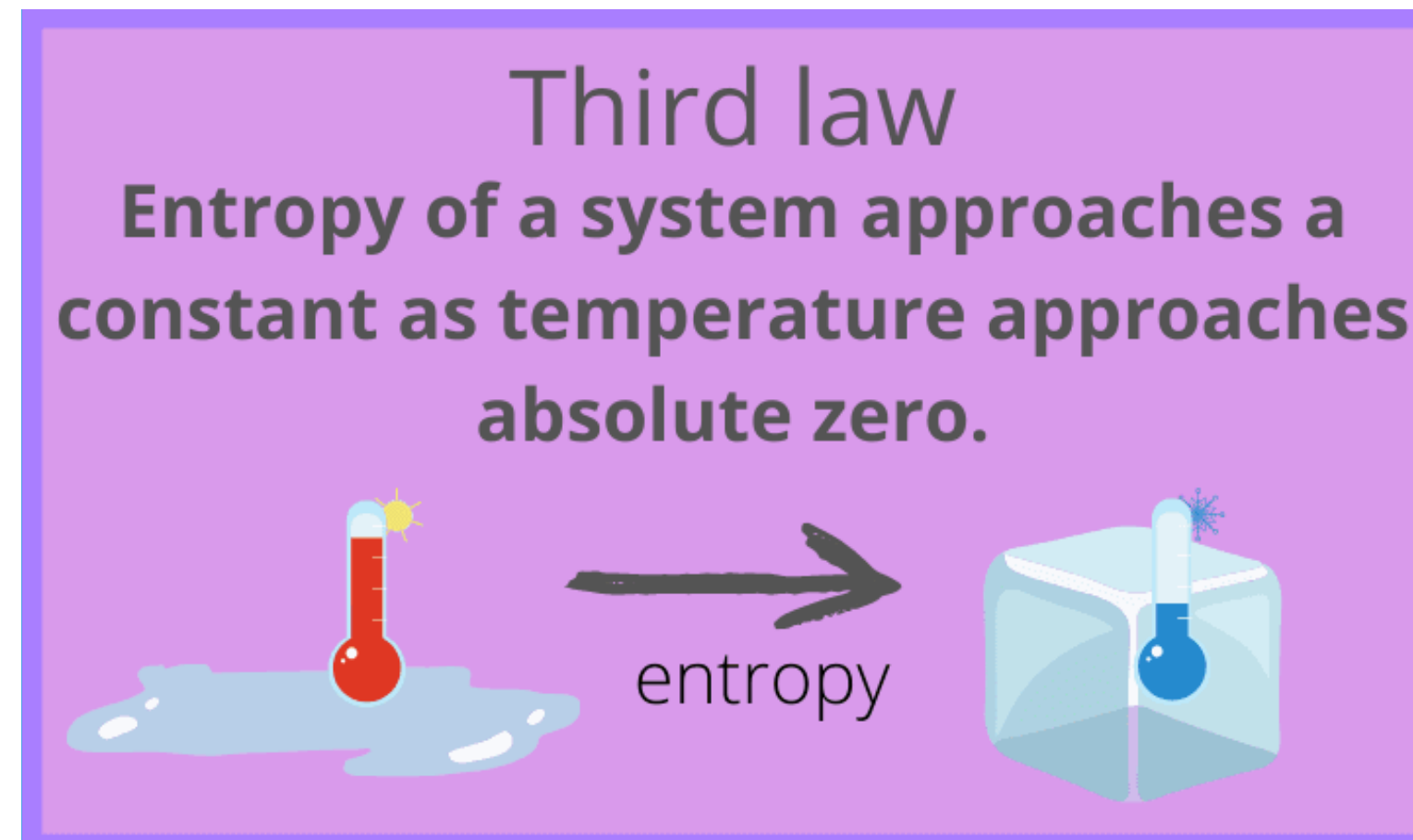
It implies that energy conversions are never 100% efficient, such as in combustion engines or drying systems where some energy is lost as heat.



Third Law of Thermodynamics

The Third Law explains that at absolute zero (0K), a perfect crystalline substance has minimal entropy.

While this is a theoretical concept, it sets a baseline for entropy calculations in chemical and biological processes in biosystems.



P-v Diagram

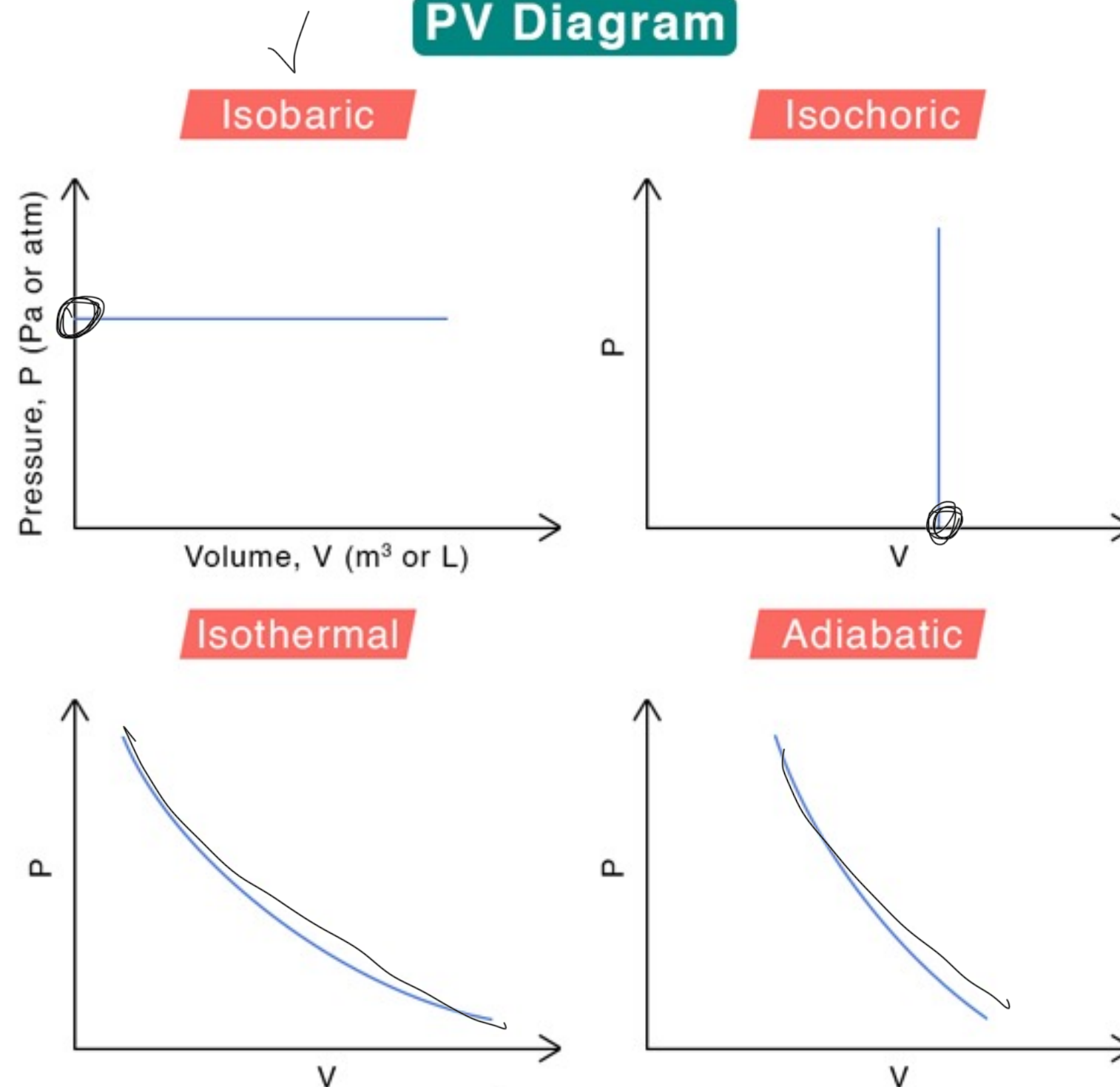
A pressure-volume (P-v) diagram is a graphical way of representing the relationship between the pressure and volume of an ideal gas.

A P-v diagram is drawn for a thermodynamic process occurring in a closed system, where there is no exchange of matter and energy between the system and its surroundings.

The relationship between pressure and volume is causal, meaning a change in one results in a change in the other.

Types of P-v Diagrams

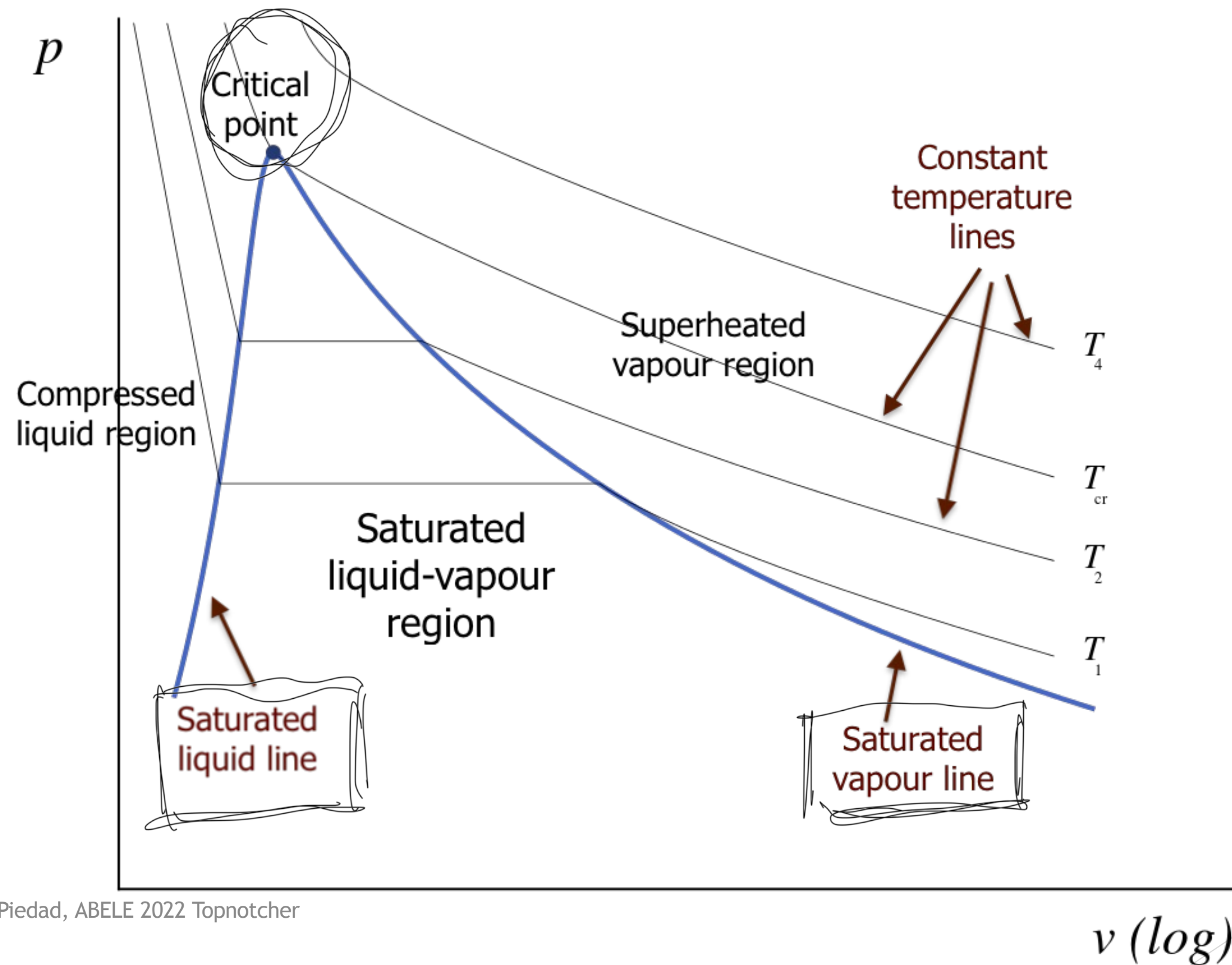
PV Diagram



1. **Isobaric** – when the gas is held at a constant pressure
2. **Isochoric** – when the gas is held at a constant volume
3. **Isothermal** – when the gas is held at a constant temperature
4. **Adiabatic** – no heat flows in and out of the container

Source: <https://www.sciencefacts.net/pv-diagram.html>

Types of P-v Diagrams



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Work done from P-v Diagram

Using P-v Diagram, we can calculate the change in internal energy, heat exchanged between the system and the surroundings, and work done. To determine work, we must integrate and calculate the area under the PV curve.

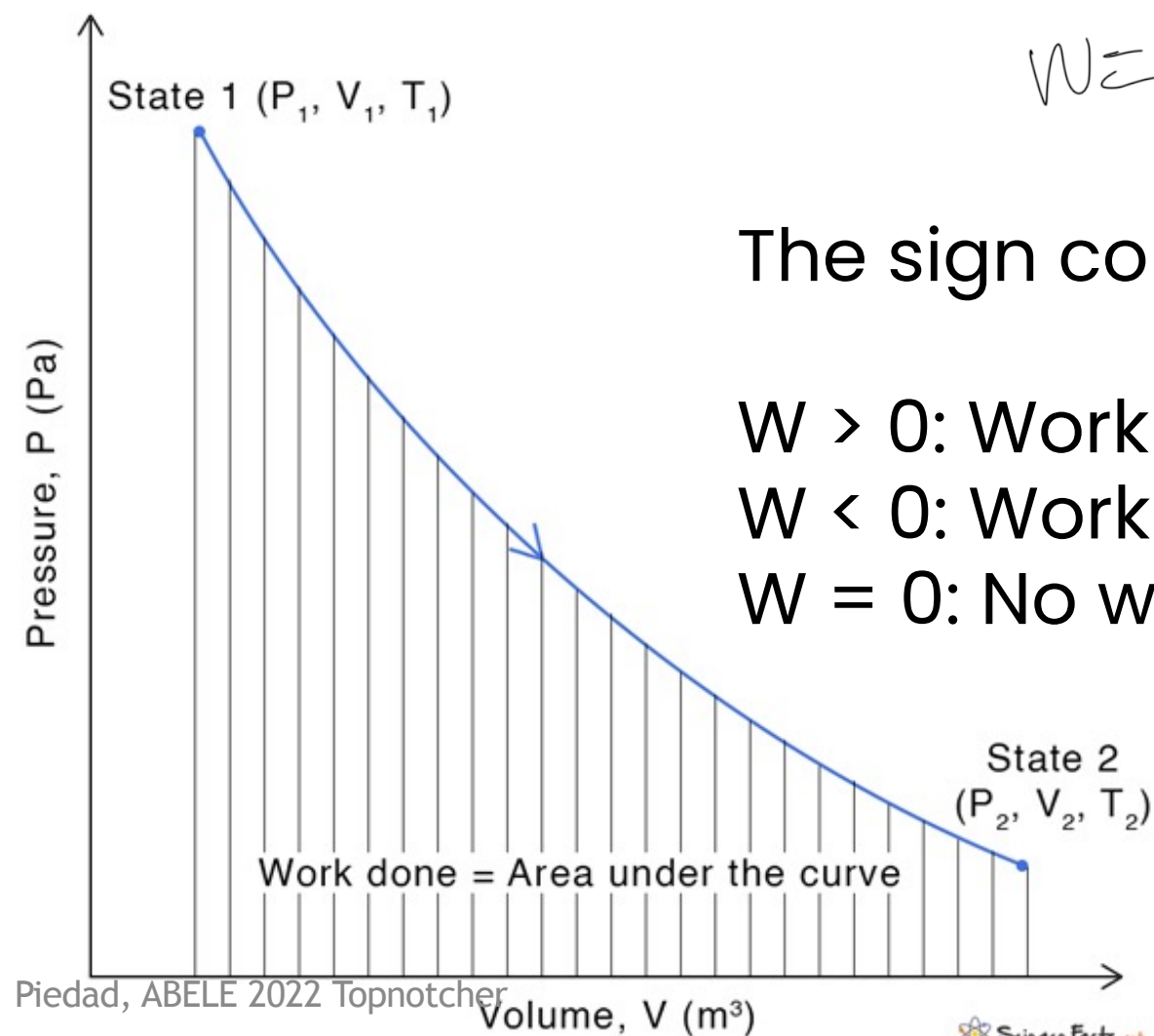
$$W = \int P dV$$

The sign convention used for W is as follows:

$W > 0$: Work is done on the system or compression

$W < 0$: Work is done by the system or expansion

$W = 0$: No work is done or isochoric process



P-T Diagram

A pure substance may exist in any of the three phases: solid, liquid, and vapour, at certain temperatures and pressures. When its temperature or pressure changes, a substance may transition from one phase to another. For example, liquid water at 1 atm turns into ice when its temperature drops to the freezing point of 0°C.

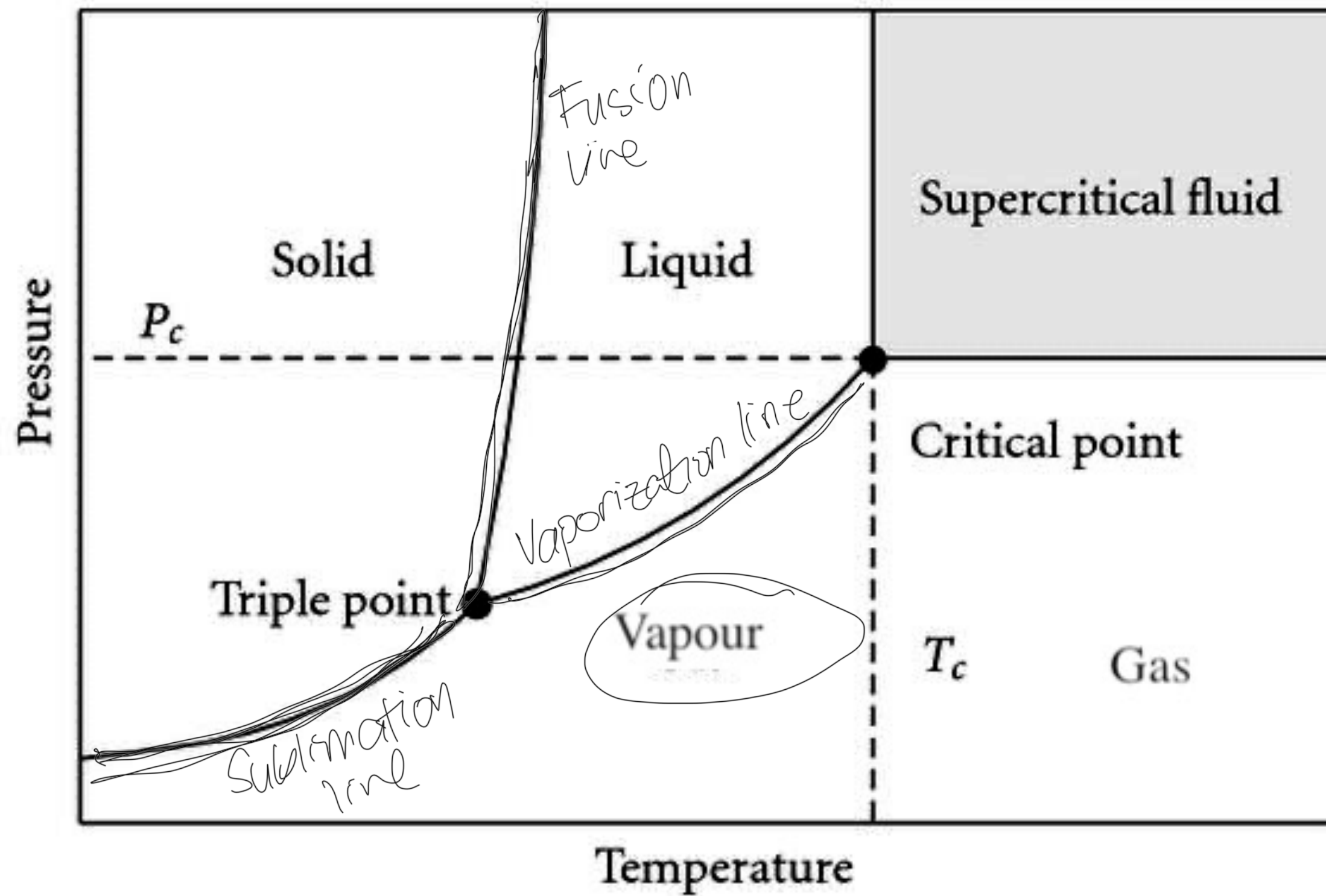
Vaporization Line – curve that lies between the liquid and vapour phases

Fusion line – curve that represents the transition between the solid and liquid phases

Sublimation line – curve below the triple point is called sublimation line, across which a substance can change directly from solid to vapour or vice versa without a transition through the liquid phase

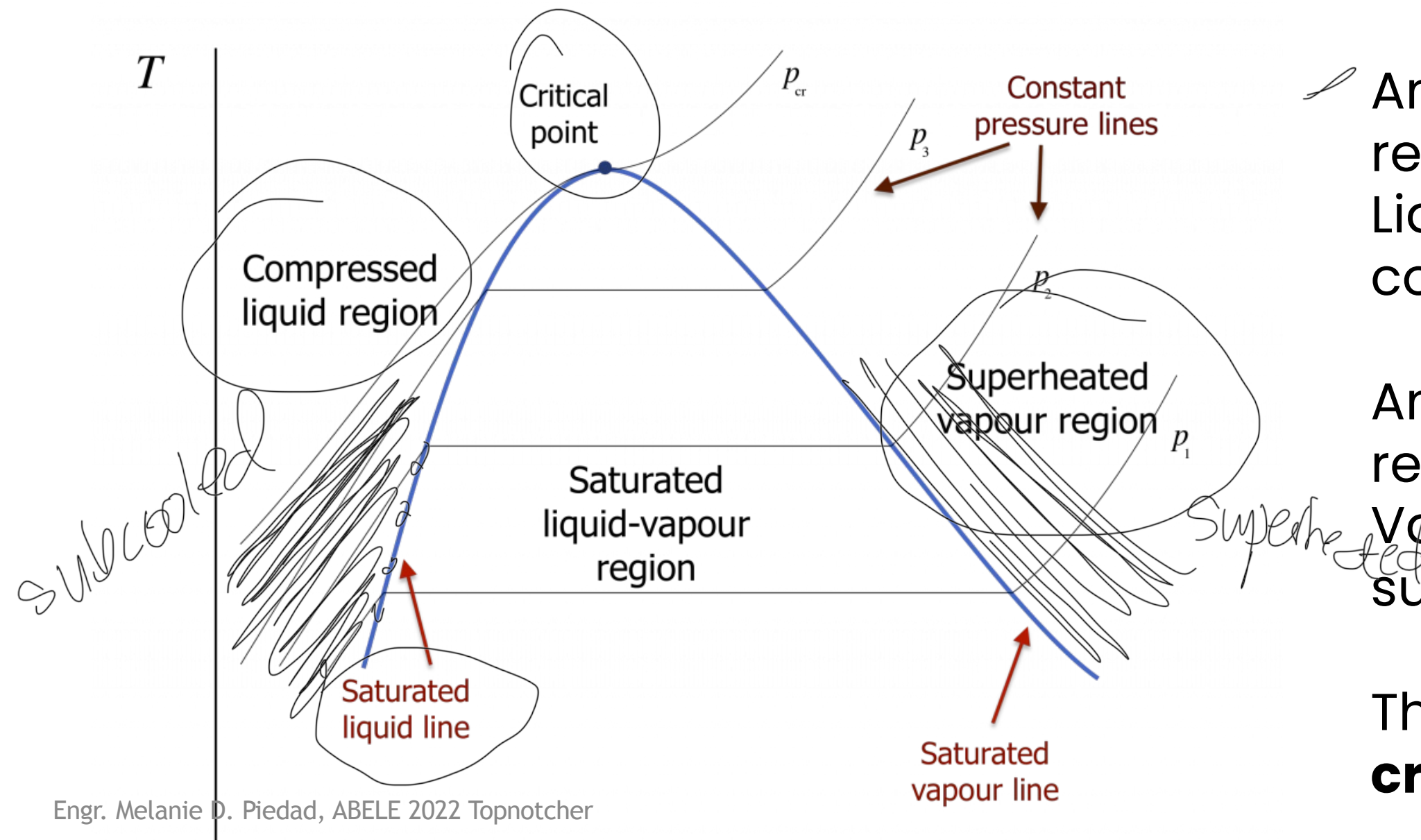
Triple point – point where vaporization, fusion and sublimation lines meet

P-T Diagram



T-v and P-v Diagram

In many thermodynamic cycles, a working fluid experiences phase changes between liquid and vapour in the subcritical zone (e.g., water in steam power plant, R134a in vapour-compression refrigeration system).



Any point on the saturated liquid line represents a **saturated liquid state**. Liquid state is commonly called compressed liquid or subcooled liquid.

Any point on the saturated vapour line represents a **saturated vapour state**. Vapour state is commonly called superheated vapour.

The two saturation lines meet at the **critical point**.

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States

State	Phase	Condition	Example
Saturated Liquid	100% liquid	At boiling point (about to vaporize)	Water at 100C, 1 atm
Saturated Vapor	100% vapor	At condensation point (about to condense)	Steam at 100C, 1 atm
Saturated Mixture	Liquid + Vapor	Exists between saturated liquid and saturated vapor	Boiling water
Subcooled Liquid	100% liquid	Temperature $<$ saturation temperature at that pressure	Water at 50C, 1 atm
Superheated Vapor	100% vapor	Temperature $>$ saturation temperature at that pressure	Water at 150C, 1 atm

Mollier Diagram

The Mollier diagram is a graph used in thermodynamics to visualize the relationships between temperature, pressure, specific volume, enthalpy, and entropy of a substance. It's also known as the **enthalpy-entropy chart**.

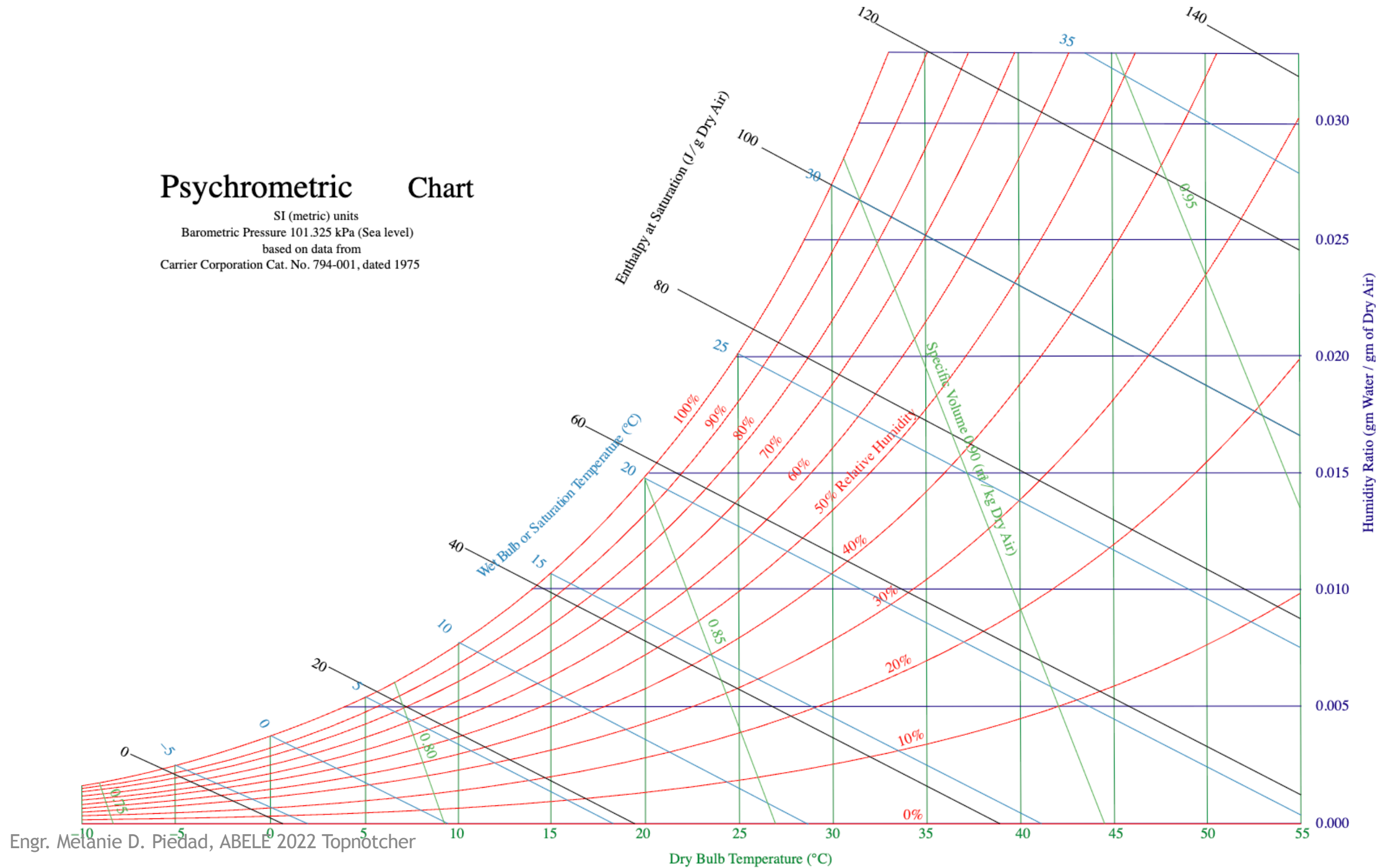
This diagram is used to visualize the operating cycles of thermodynamic systems when designing power plants, compressors, steam turbines, refrigeration systems, and air conditioning equipment by illustrating cooling, heating, or even humidification processes.

This diagram is also known as the **psychrometric chart**.

Mollier Diagram

Psychrometric Chart

SI (metric) units
Barometric Pressure 101.325 kPa (Sea level)
based on data from
Carrier Corporation Cat. No. 794-001, dated 1975



Phase Changes

Phase Change	Name of Process	Energy Behavior
Solid to liquid	Melting (fusion)	Absorbs energy (latent heat of fusion)
Liquid to vapor	Vaporization (boiling)	Absorbs energy (latent heat of vaporization)
Vapor to liquid	Condensation	Releases energy
Liquid to solid	Freezing	Releases energy
Solid to vapor	Sublimation	Absorbs energy
Vapor to solid	Deposition	Releases energy

Quality (x)

Quality, symbolized by x , refers to the mass fraction of vapor in a saturated liquid-vapor mixture.

It only applies in the two-phase (saturation) region — between saturated liquid and saturated vapor.

$$x = \frac{m_{\text{vapor}}}{m_{\text{total}}} = \frac{h - h_f}{h_{fg}}$$

Where:

x = quality ($0 \leq x \leq 1$)

h = specific enthalpy of the mixture

h_f = enthalpy of saturated liquid

$h_{fg} = h_g - h_f$ = latent heat of vaporization

$x = 0.65$
65% saturated vapor
35% saturated liquid
liquid-vapor = 1
0.65

Quality (x)

Value of x	Meaning
$x = 0$	Saturated liquid (no vapor)
$0 < x < 1$	Mixture of liquid and vapor
$x = 1$	Saturated vapor (no liquid)
$x > 1$ or $x < 0$	Not physically valid in two phase region

Enthalpy

Enthalpy (H) is the sum of internal energy and the product of pressure and volume.

It's used in **drying calculations, heating systems, and determining energy requirements in food and biomass processing.**

Sample Question

A water-steam mixture has an enthalpy of 2100 kJ/kg. At 101.3 kPa (1 atm), the enthalpy of saturated liquid is 419 kJ/kg and latent heat of vaporization is 2257 kJ/kg.

A. 0.79

B. 0.64

C. 0.74

D. 0.69

$$h = h_f + x h_{fg}$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
h of mixture h of saturation quality h of vaporization

$$2100 \frac{\text{kJ}}{\text{kg}} = 419 + x (2257)$$

$$x = 0.74$$

74% saturated vapor
26% saturated liquid

Types of Enthalpy

Enthalpy of fusion is the amount of thermal energy (usually in kJ/mol) required to change a substance from the **solid phase to the liquid** phase at its melting point without changing its temperature.

This is crucial in phase change processes like melting ice into water, which is highly relevant in thermal storage, food freezing/thawing systems, and phase change materials in ABE.

Types of Enthalpy

Enthalpy of sublimation is the energy required to convert a solid directly into a gas, skipping the liquid phase.

Example: Dry ice (solid CO_2) sublimating into CO_2 gas. //

Enthalpy of liquefaction is the energy released (not required) when a gas turns into a liquid. Technically, it's the negative of enthalpy of vaporization. Not commonly used terminology — the more accurate and accepted term is enthalpy of condensation.

Enthalpy of vaporization is the energy required to convert a liquid into a gas at its boiling point. It's important in drying, boiling, and distillation processes in food engineering.

Enthalpy

Enthalpy is the sum of internal energy and flow work.

$$H = U + Pv$$

Open and Closed Systems

The first law for closed systems accounts for energy conservation with work and heat.

On the other hand, open systems include mass flow, so energy includes enthalpy, kinetic, potential, and shaft work.

Where:

ΔU = change in internal energy

Q = heat added to the system

W = work done by the system

Work and Heat Interactions

Boundary work results from volume change under pressure, such as piston movement.

Common in pistons, gas chambers, and engines.

$$W_{\text{boundary}} = \int_{v1}^{v2} P dV$$

$$W = \int P dv$$

For a constant pressure process:

$$W = P(V_2 - V_1)$$

Work and Heat Interactions

Shaft work is mechanical work done by rotating devices like turbines or pumps.

$$W_{\text{shaft}} = 2\pi nT$$

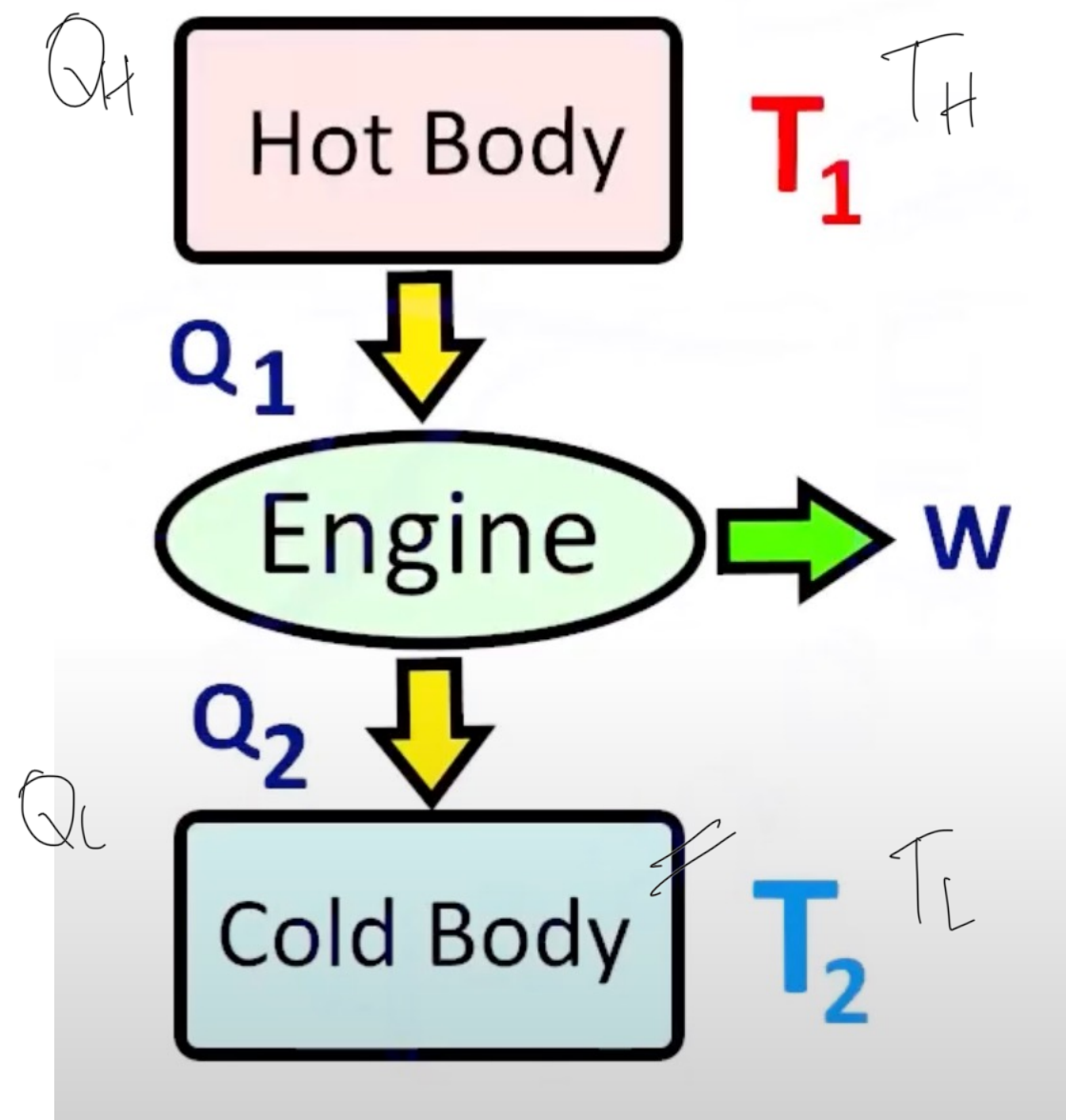
Where:

n=number of revolutions

T=torque

Heat Engine

A heat engine converts thermal energy into mechanical work.



Q_1 = Heat added

W = Work output (done by the engine) = $(Q_2 - Q_1)$

Q_2 = Heat of exhaust gases (heat rejected)

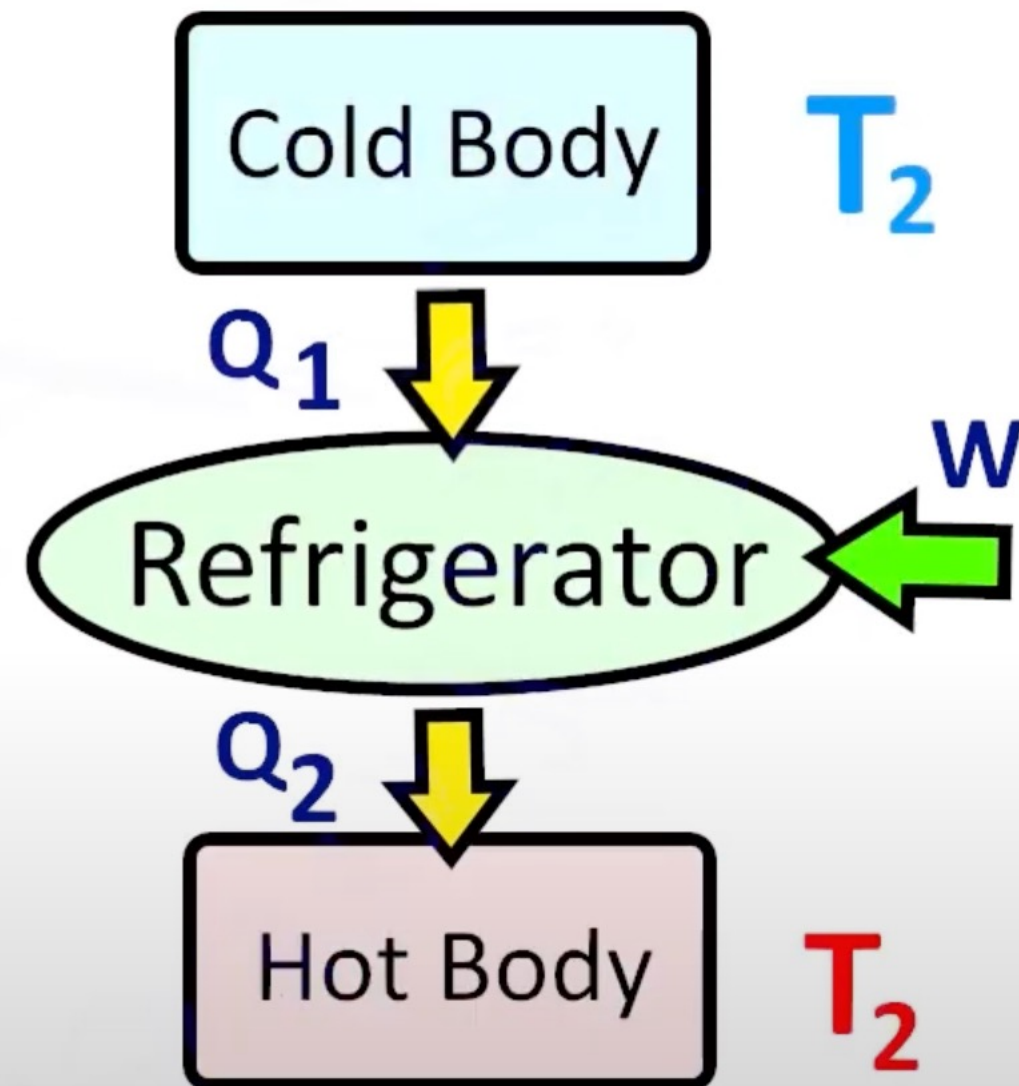
Efficiency of engine: $\mu = \frac{\text{output}}{\text{input}} = \frac{Q_1 - Q_2}{Q_1}$

Example:

1. Steam engine running on biomass for a rice mill.
2. Biogas generator in farms converting manure to power

Refrigerators

A refrigerator extracts heat from a low-temperature region and discharges it to a higher-temperature region.



Q_1 = Heat removed

W = Work input (done on the refrigerator) = $(Q_2 - Q_1)$

Q_2 = Heat delivered

COP of refrigerator:
$$\text{COP} = \frac{\text{Cooling effect}}{\text{input}} = \frac{Q_1}{Q_2 - Q_1}$$

Example:

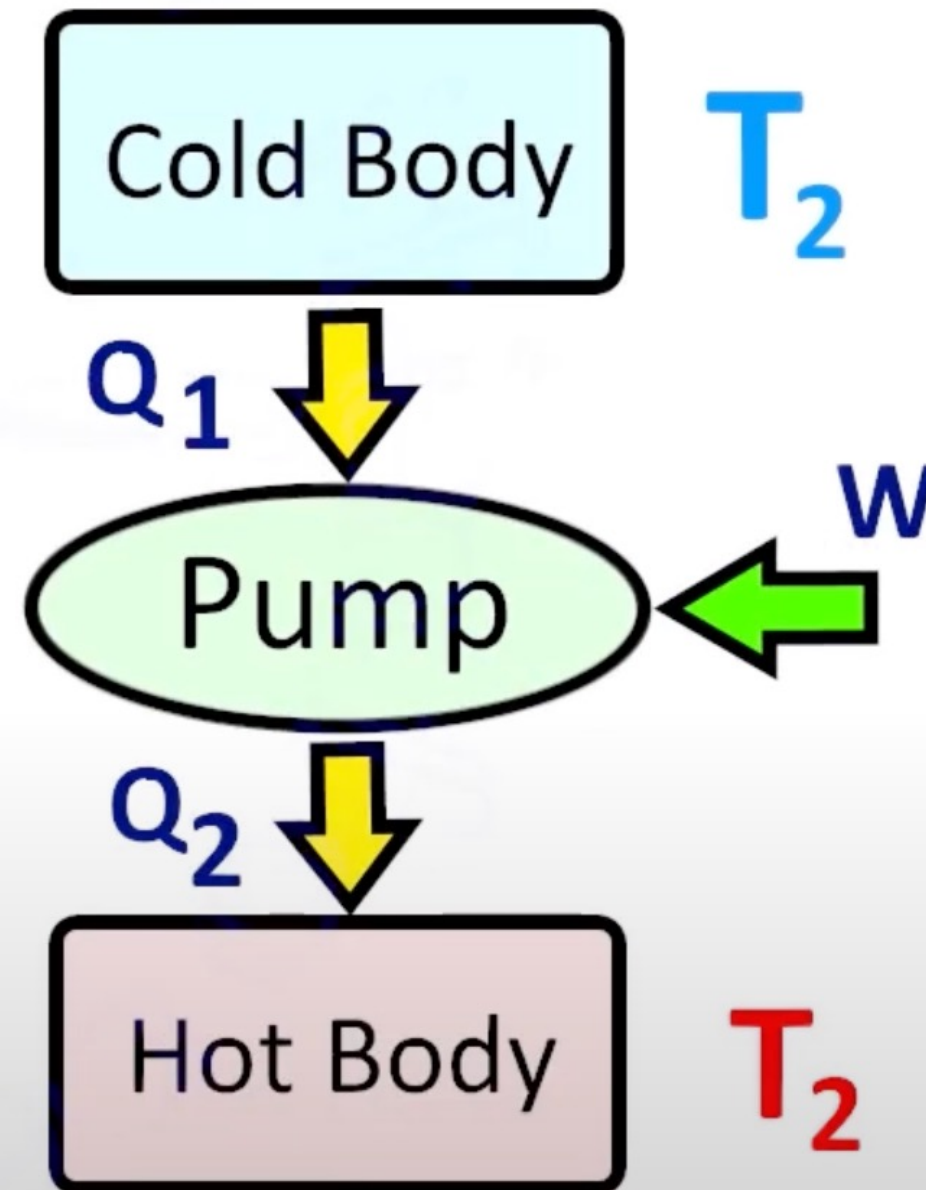
1. Cold storage for fruits, vegetables, or meat.
2. Milk chilling systems in dairy farms.

Heat Pumps

Q_1 = Heat removed

W = Work input (done on heat pump) = $(Q_2 - Q_1)$

Q_2 = Heat delivered



COP of refrigerator: $\text{COP} = \frac{\text{Heating effect}}{\text{input}} = \frac{Q_2}{Q_2 - Q_1}$

COP of heat pump is always greater than 1.

COP of Heat Pump = 1 + COP of refrigerator

COP of Heat Pump = $1 + \frac{Q_1}{Q_2 - Q_1}$

Example:

1. Heating greenhouses in cold climates.
2. Ground-source heat pumps for indoor farming setups.

Sample Question

A biomass-powered heat engine receives 3,000 kJ of heat from a high-temperature source and rejects 1,200 kJ to the surroundings. Calculate the work output and thermal efficiency.

A. 4,200 kJ, 60%

B. 1,800 kJ, 67%

C. 1,800 kJ, 60%

D. 4,200 kJ, 67%

$$\begin{aligned}\text{Work output} &= 3000 - 1200 \\ &= 1800 \text{ kJ}\end{aligned}$$

$$\eta = \left(\frac{1800}{3000} \right) (100) = 60\%$$

Sample Question

A diesel engine operating in a rice farm runs as a heat engine receiving 10,000 kJ of energy and operates at 35% efficiency. Calculate the amount of heat rejected.

A. 3,500 kJ

B. 6,500 kJ

C. 8,775 kJ

D. 2,500 kJ

$$W = (35\%)(10,000) = 3500 \text{ kJ}$$

$$W = Q_A - Q_{Rj}$$
$$10000 - 3500$$

$$Q_{Rj} = 6500 \text{ kJ}$$

Sample Question

A dairy refrigerator removes 500 kJ of heat from milk and requires 120 kJ of work. What is the coefficient of performance of the refrigerator?

$$\text{COP} = \frac{Q_A}{W} = \frac{500}{120}$$

A. 4.17

B. 0.24

C. 4.0

D. 3.15

Sample Question

A refrigeration system used for cold storage extracts 750 kJ of heat per cycle with a COP of 3.5. Determine the amount of heat rejected to the surroundings.

$$\text{COP} = \frac{Q_A}{W} = \frac{Q_A}{Q_R - Q_A}$$

$$3.5 = \frac{750}{Q_R - 750}$$

$$Q_R = 964.29 \text{ kJ}$$

A. 214.29 kJ

B. 750 kJ

C. 262.5 kJ

D. 964.29 kJ

Sample Question

A heat pump supplies 1,200 kJ of heat to an aquaculture pond and operates with a COP of 4. Determine the amount of heat extracted from the cold reservoir.

- A. 300 kJ
- B. 900 kJ
- C. 600 kJ
- D. 1,200 kJ

$$COP = \frac{Q_R}{Q_R - Q_A}$$

heating effect

$$COP = \frac{Q_A}{Q_R - Q_A}$$

cooling effect

↓
heat pump

↓
refrigerator

$$4 = \frac{1200 \text{ kJ}}{1200 - Q_A}$$

$$Q_A = 900 \text{ kJ}$$

Kevin Planck Statement

Kevin Planck statement about heat engines:

"It is impossible for any device that operates on a cycle to receive heat from a single thermal reservoir and produce a net amount of work."

It means that:

- A heat engine cannot convert all the heat it receives into work.
- Some energy must be rejected to a cooler reservoir.
- No 100% efficient heat engine is possible.

Violating example: A machine that takes heat from a furnace and turns all of it into electricity without waste heat – impossible.

Rudolf Clausius Statement

Rudolf Clausius statement about refrigerators and heat pumps:

"It is impossible to construct a device that operates in a cycle and transfers heat from a cooler body to a hotter body without external work."

It means that:

- Heat cannot flow from cold to hot by itself.
- A refrigerator or heat pump requires input work to move heat against the natural flow.
- You can't build a refrigerator that cools a space without using energy.

Violating example: A refrigerator that cools produce in a farm storage room without any electricity — impossible.

Irreversibility

Mixing increases entropy and cannot be reversed without external work.

The system and its environment cannot be restored to their original states at the same time. The sign of an irreversible process comes from the finite gradient between the states occurring in the actual process.

For example, when heat flows from one object to another, there is a finite temperature difference (gradient) between the two objects. More importantly, at any given moment of the process, the system most likely is not at equilibrium or in a well-defined state. This phenomenon is called **irreversibility**.

Reversibility

A reversible process is a process in which the system and environment can be restored to exactly the same initial states that they were in before the process occurred, if we go backward along the path of the process.

The necessary condition for a reversible process is therefore the quasi-static requirement. Note that it is quite easy to restore a system to its original state; the hard part is to have its environment restored to its original state at the same time.

For example, in the example of an ideal gas expanding into vacuum to twice its original volume, we can easily push it back with a piston and restore its temperature and pressure by removing some heat from the gas. The problem is that we cannot do it without changing something in its surroundings such as dumping some heat there.

Carnot Cycle

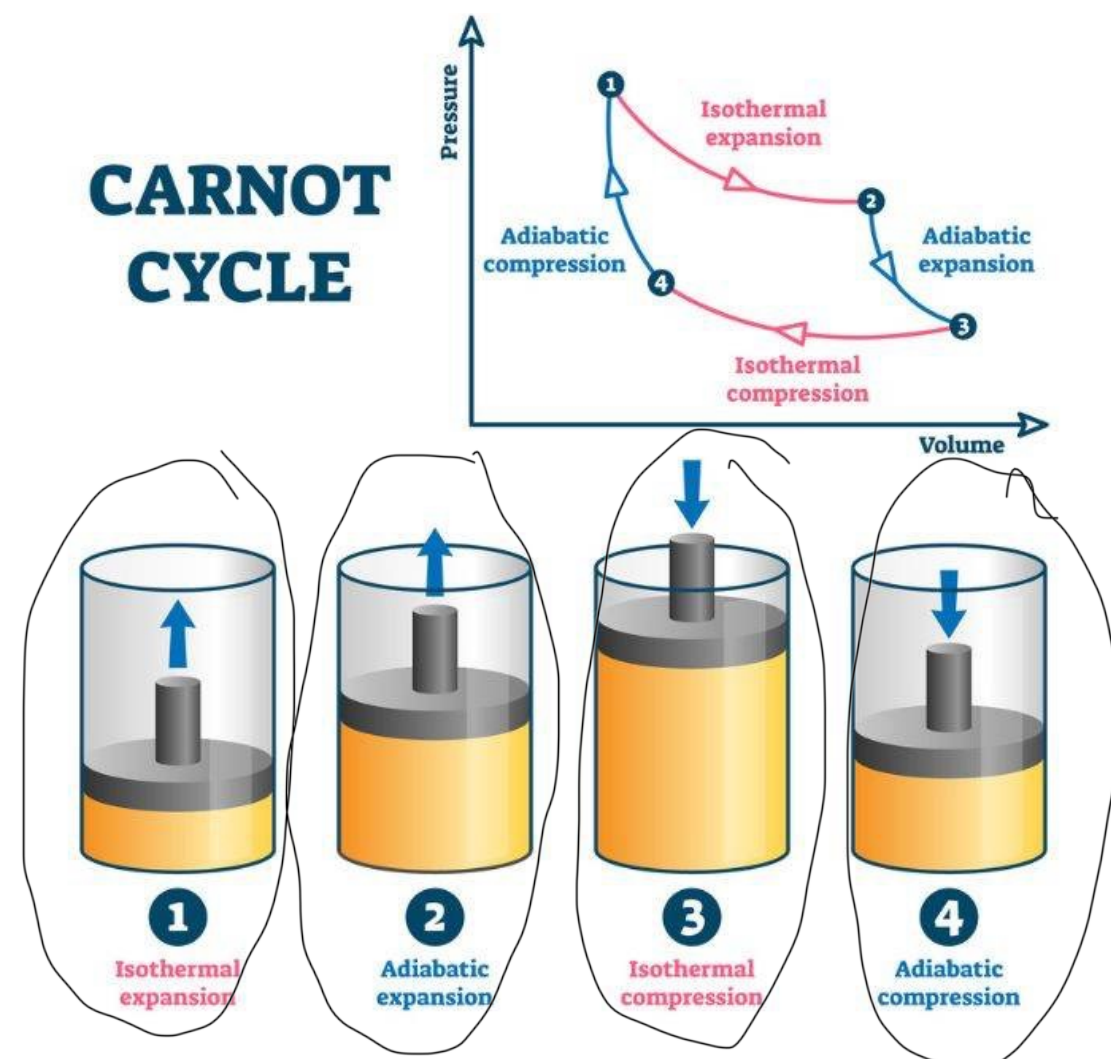
Carnot efficiency depends only on **high and low reservoir temperatures**.

The Carnot cycle is a theoretical thermodynamic cycle proposed by Sadi Carnot in 1824. It represents an idealized heat engine that operates reversibly between two thermal reservoirs:

- A hot reservoir at temperature T_H
- A cold reservoir at temperature T_L

This is the **most effective** cycle known to exist.

Carnot Cycle



Carnot efficiency:

$$\mu_{\text{carnot}} = 1 - \frac{T_L}{T_H}$$

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Isothermal Expansion: Heat is absorbed from the hot reservoir at constant temperature T_H ; gas expands and does work.

Adiabatic Expansion: The system expands without heat transfer, and its temperature decreases from T_H to T_L .

Isothermal ^{Compression} Expansion: The system rejects heat to the cold reservoir at T_L while being compressed.

Adiabatic Compression: The system is compressed without heat exchange, raising its temperature back to T_H .

Sample Question

~~What determines the efficiency of a Carnot engine?~~ A Carnot engine operates between a hot reservoir at 500 K and a cold reservoir at 300 K. What is its efficiency?

A. 20%

B. 30%

C. 40%

D. 50%

$$\eta = \left[1 - \frac{300}{500} \right] (100)$$

$$\eta = 40\%$$

Entropy

Entropy (S) is a thermodynamic property that measures:

- The degree of disorder or randomness in a system.
- The dispersion or spreading of energy.
- The unavailability of energy to do useful work.

Entropy

In any real process, the total entropy of the system and its surroundings **always increases.**

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$$

If:

= 0 ; reversible process

> 0 ; irreversible process

Interpretation:

- Nature favors disorder.
- Heat naturally flows from hot to cold.
- Friction, mixing, and spontaneous processes all cause

Rankine Cycle

The fundamental thermodynamic cycle used in all steam power plants is the Rankine cycle. Other applications, such as solar thermal power plants and geothermal power plants, also make use of it.

The Rankine cycle is a thermodynamically reversible cycle, which means that all of the heat input may be converted back into labor. Friction, heat transmission, and other variables cause losses in practice nevertheless. As a result, the Rankine cycle's efficiency is normally between 30 and 40%.

The Rankine cycle is available in a variety of forms, including the supercritical, reheat, and regenerative cycles. Through a reduction in the quantity of heat rejected by the environment, these modifications aim to increase the cycle's efficiency.

Rankine Cycle

Rankine Cycle Modification	Purpose	Key Effect
Reheat cycle	Avoid turbine moisture, improve efficiency	Adds another expansion stage
Regenerative cycle	Save energy by preheating feedwater	Reduces Q required in boiler
Binary cycle	Use low-grade heat sources	Adapts for low-temp energy recovery

Otto Cycle

Many modern internal combustion engines still use the Otto cycle.

ICE

The Otto cycle is a simplified thermodynamic cycle that depicts how a common spark ignition piston engine works. It is the Highest frequent thermodynamic cycle seen in car engines.

The Otto cycle is an imagined cycle, although actual engines do not follow it exactly. The Otto cycle, on the other hand, is a valuable model for understanding how spark ignition engines work.

Note: Gasoline engines (Spark-ignition)

Diesel Cycle

Diesel cycle is more effective than Otto cycle. This is so that the gasoline may ignite more readily and completely. A diesel engine's increased compression ratio raises the temperature at the end of the compression stroke. Diesel engines are therefore more efficient than petrol engines.

Additionally, the Diesel cycle is stronger than the Otto cycle. This is due to the fact that a spark plug is not used to ignite the fuel in a diesel engine, thus there is no chance of a misfire if the spark plug breaks. As a result, diesel engines are frequently utilized in settings where dependability is crucial, such as heavy-duty vehicles and buses.

Note: Diesel engines (Compression-ignition)

Efficiency in Thermodynamics

Thermal efficiency helps determine how effectively a system converts input energy (fuel, electricity, etc.) into useful work or heat.

In ABE, it's essential for assessing the **performance of biomass stoves, crop dryers, irrigation pumps, and heat recovery systems. adiabatic process involves no heat transfer into or out of the system.**

Ideal Gas Law

The Ideal Gas Law: $PV=nRT$  $PV = nRT$

P=pressure (atm, mmHg, torr)

V=volume (L)

n=amount of substance (moles)

R=ideal gas constant

T=temperature (Kelvin)

It is essential for **estimating gas behavior in closed systems like biogas digesters or sealed dryers.**

Ideal Gas Law

R is a constant that links energy units with pressure, volume, and temperature. Its value depends on the units used.

Numerical Value of R	Unit
0.08205	$\frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$
62.36	$\frac{\text{L} \cdot \text{torr}}{\text{mol} \cdot \text{K}}$ or $\frac{\text{L} \cdot \text{mmHg}}{\text{mol} \cdot \text{K}}$
8.314	$\frac{\text{J}}{\text{mol} \cdot \text{K}}$

Boyle's Law (Isothermal)

Formula: $P_1V_1 = P_2V_2$

Where:

P=pressure

V=volume

Used in sealed grain storage or vacuum systems, where gas compression and volume change are studied under fixed temperature.

Charles's Law (Isobaric)

Formula: $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

Where:

V=volume

T=temperature

Used in greenhouses, air handling, and ventilation, where air volume changes with heat.

Gay Lussac's Law (Isochoric / Isometric)

Formula: $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

Where:

P=pressure

T=temperature

It explains how pressure inside a sealed tank or chamber increases with heating – **useful in sterilizers and fermentation tanks.**

Sample Question

How much pressure is exerted by 2.3 mol of H₂ gas placed in a 5 L container at 96.0 degrees Fahrenheit?

$$PV = nRT$$

$$n = 2.3 \text{ mol H}_2$$

$$V = 5 \text{ L}$$

$$T = 96^\circ\text{F} = 35.56^\circ\text{C} + 273 = 308.56 \text{ K}$$

$$R = 0.08205 \frac{\text{L} \cdot \text{atm}}{\text{mol K}}$$

$$(P)(5\cancel{\text{L}}) = (2.3\cancel{\text{mol}}) \left(0.08205 \frac{\cancel{\text{L}} \cdot \text{atm}}{\cancel{\text{mol}} \cancel{\text{K}}} \right) (308.56\cancel{\text{K}})$$

$$P = 11.65 \text{ atm}$$

Sample Question

A gas occupies 4.5 liters at a pressure of 60 mmHg. What is the volume when the pressure is increased to 90 mmHg?

Boyle's Law: $P_1 V_1 = P_2 V_2$

$$(60 \text{ mmHg})(4.5 \text{ L}) = (90 \text{ mmHg})(V_2)$$

$$V_2 = 3 \text{ L}$$

Sample Question

To what pressure must a gas be compressed in order to get into an 8 cu. ft. tank the entire weight of a gas that occupies 68 cu. ft. at standard pressure?

$$P_1 V_1 = P_2 V_2$$

$$(1 \text{ atm})(68 \text{ cu. ft.}) = (P_2)(8 \text{ cu ft.})$$

$$P_2 = 8.5 \text{ atm}$$

Sample Question

Originally, a sample of $\text{CO}_2(\text{g})$ is in a rigid container at 289 K and 0.75 atm. What is the new pressure when the $\text{CO}_2(\text{g})$ in the container is heated to 435 K?

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \quad \text{Charles's Law}$$

$$\frac{(0.75 \text{ atm})}{(289 \text{ K})} = \frac{(P_2)}{(435 \text{ K})}$$

$$P_2 = (1.1 \text{ atm}) \left(\frac{101,325 \text{ Pa}}{1 \text{ atm}} \right) = 111,457.5 \text{ Pa}$$

A. 1.1 atm

B. 1.2 atm

C. 1.0 atm

D. 1.3 atm

Sample Question

Calculate the decrease in temperature (in Celsius) when 2.00 L at 21.0°C is compressed to 1.00 L.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Charles's law

$$\frac{2\text{ L}}{21} = \frac{1\text{ L}}{t_2}$$

A. 10.5 °C

B. 11.1 °C

C. -126 °C

D. -105 °C

Sample Question

What change in volume results if 76 mL of gas is cooled from 65 °C to 12 °C?

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{(76 \text{ mL})}{(65 + 273)} = \frac{(V_2)}{(12 + 273)} \rightarrow V_2 = 64.08 \text{ mL}$$

$$\Delta V = 76 - 64.08$$

$$\Delta V = 11.92 \text{ mL}$$

Sample Question

An automobile tire is inflated to 43 psig pressure at 70°F. After being driven, the temperature rises to 95°F. Assuming that the volume remains constant, the final gage pressure is:

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \quad \text{Gay-Lussac's Law}$$
$$\frac{(43 + 14.7)}{(70 + 460)} = \frac{P_2}{(95 + 460)} \rightarrow P_2 = 60.42$$
$$P_2 \text{ in gage} = 45.72 \text{ psig}$$

Sample Question

A sample of gas at 5.00×10^3 mmHg inside a steel tank is cooled from 200.0°C to 20.0°C . What is the final pressure of the gas in the steel tank in KPa?

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{(5 \times 10^3 \text{ mmHg} \times \frac{101.325 \text{ kPa}}{760 \text{ mmHg}})}{(200 + 273)} = \frac{P_2}{(20 + 273)}$$

$$P_2 = 412.93 \text{ kPa}$$

Sample Question

The atmospheric pressures at the top and bottom of a building are read by a barometer to be 86.0 and 88.0 kPa. What is the height of the building?

$$-\Delta P = \rho g \Delta h$$

$$-(86,000 - 88,000) = \left(1.2 \frac{\text{kg}}{\text{m}^3}\right) \left(9.81 \frac{\text{m}}{\text{s}^2}\right) \Delta h$$

$$\Delta h = 169.89 \approx 170 \text{ m}$$

Sample Question

A 8-m³ rigid vessel contains steam at 30 MPa and 300°C. The mass of the steam is:

Sample Question

Air is heated in a fixed container from 300K to 600K. What happens to the pressure?

A. It halves

B. It doubles

C. It stays the same

D. It quadruples

$$\begin{aligned} \frac{P_1}{T_1} &= \frac{P_2}{T_2} \\ 1 \frac{P_1}{300} &= \frac{P_2}{600} \\ \frac{1}{300} &= \frac{P_2}{600} \\ \frac{600}{300} &= P_2 \end{aligned} \quad \rightarrow \quad \textcircled{2} P_2 \text{ doubles}$$